

Empirical analytics: inputs and outputs of the model

The model is a belief over two binary attributes of a candidate, competence (A) and trustworthiness (B), updated by two soft cues, a credential and a reference letter. The attributes are discrete. Every belief about them is described by continuous parameters, and the results are expansions in one of those parameters, the prior covariance c . The decision is discrete again, and becomes smooth in c only through the population density f of the net surplus u . Table 1 lists the inputs and Table 2 the outputs. The last column separates what is elicited from an evaluator, what is recorded by the observer, and what can only be computed.

Table 1: Inputs of the model, their mathematical type, and what each represents. The attributes are discrete; every belief about them is described by continuous parameters.

Symbol	Type	Represents	Obtained how
$A, B \in \{0, 1\}$	discrete variables	whether the candidate is competent ($A=1$) and trustworthy ($B=1$)	never observed; the objects of belief
$P = (P_{ij})$	probability mass function on the four pairs (i, j)	the evaluator's belief, before any cue, about which traits a randomly drawn candidate has; no outcome is involved	elicited before the file is opened
α, β	probabilities in $(0, 1)$; $(\alpha, 1 - \alpha)$ is the A -marginal of P , $(\beta, 1 - \beta)$ its B -marginal	the believed base rate of candidates who <i>lack</i> competence, $\alpha = P(A=0)$, and who lack trustworthiness, $\beta = P(B=0)$	one base-rate question per trait
$c = \text{assoc}(P)$	real number within the Fréchet–Hoeffding bounds; $c = 0$ is independence	how strongly the evaluator believes, before any cue, that the two traits go together	a joint question before the cues
$q = (q_0, q_1)$	probabilities, $q_0 + q_1 = 1$, $0 < q_0 < 1$; replaces the A -marginal $(\alpha, 1 - \alpha)$	the credence that the candidate is not competent (q_0) or competent (q_1) after the credential alone	a rating of competence, on a probability scale, after the credential alone
$r = (r_0, r_1)$	probabilities, $r_0 + r_1 = 1$, $0 < r_0 < 1$; replaces the B -marginal $(\beta, 1 - \beta)$	the credence that the candidate is not trustworthy (r_0) or trustworthy (r_1) after the letter alone	a rating of trustworthiness after the letter alone
$\sigma \in \{AB, BA\}$	discrete label	which of the two documents the evaluator read first	recorded, or assigned by the experimenter
λ	fraction in $[0, 1]$	the share of evaluators who read the credential first	counted where σ is recorded; unknown in a pooled audit
$v = (v_{ij})$	four real numbers, not probabilities	what a candidate of each trait combination is worth to the evaluator; known to the evaluator and never updated by a cue. It may embed the chance that such a candidate succeeds, which the model does not separate out	stated payoffs, or inferred from choices
τ	real number	the value of the outside option forgone by engaging	as for v ; enters only through u
$u = s(P^B) - \tau$, density f	real number for each evaluator; the one input that enters as a distribution across evaluators	how far an evaluator is from indifference at the benchmark belief; $f(0)$ is the density of evaluators at the margin	not observed directly, since it depends on P^B
ε	positive real number	the resolution of the observer's instrument	set by the study design and the sample
ω	weight in $[0, 1]$	how far the later impression displaces what the earlier cue implied about its attribute; $\omega = 1$ in Sections 4 and 5	$\omega = \frac{P^A(B=1) - P_{AB}^\omega(B=1)}{P^A(B=1) - r_1}$, of the distance between the trustworthiness rating after the credential alone and the rating the letter delivers, the share that the final rating travels; requires $P^A(B=1) \neq r_1$ (Proposition ADJ)

Table 2: Outputs of the model, their mathematical type, and what each represents.

Symbol	Type	Represents	Obtained how
P_{AB}^J, P_{BA}^J	probability mass functions on the four pairs	the evaluator's belief about the candidate after both documents, in the given reading sequence	elicited after both cues, by reading group
$P_\sigma^J(A=1), P_\sigma^J(B=1)$	probabilities	the final rating of competence and of trustworthiness	one single-trait question after the cues
$\text{assoc}(P_\sigma^J)$; odds ratio, Yule's Q	real numbers	how strongly the traits are believed to go together after the cues, which is what a belief audit about traits going together measures	a joint question after the cues
$P_{AB}^J - P_{BA}^J$	signed 2×2 array summing to zero	the difference between two otherwise identical panels who read the file in opposite sequences	the two reading groups compared; needs neither P^B nor λ (Proposition ORD)
P^B	probability mass function	the belief the same evaluator would hold had the same two impressions been combined as Bayes factors	never observed; computed from P, q, r
$P_\sigma^J - P^B$	signed 2×2 array	the evaluator's departure from the sequence-free belief	computed; needs P^B
$\bar{P}_\lambda$	probability mass function	the average belief in a population that pools both reading sequences	a pooled survey
$F(\bar{P}_\lambda) - F(P^B)$	real number; protected if $\mathcal{O}(c^2)$, unprotected if first order	what a pooled audit reading the statistic F reports, relative to the benchmark	a pooled survey together with the computed P^B
$s(P) = \langle v, P \rangle$; engage iff $s(P) \geq \tau$	real number; binary decision	the expected value of engaging, and the decision to engage or not	the decision is observed, the score is not
$\mathbb{1}\{\text{sgn}(u) \neq \text{sgn}(u + c\delta_\sigma)\}$ and its population mean	binary indicator, a step function of c ; its mean is a share in $[0, 1]$	whether the reading sequence changed an evaluator's decision, and the share of evaluators for whom it did	decisions compared across reading groups
$L(c)$	nonnegative real number	the value forgone through decisions that the reading sequence changed	computed; needs u and hence P^B
$\kappa, \kappa', K, R_1, R_2, \delta_\sigma, M_\lambda$	real coefficients and 2×2 directions	none of their own: functions of α, β, q, r, v that fix the size and direction of the first-order effects above	computed